

Soham Gunturu

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EDUCATION

Georgia Institute of Technology

Expected Graduation: December 2026

GPA: 4.0 / 4.0

- *B.S. in Computer Science – Intelligence and Systems & Architecture Threads*
- Organizations: Trading at GT, AI at GT, Data Science at GT, GROWER Lab, HexLabs, Ganesh Lab, HIVE
- Coursework: Systems and Networks, Intro to AI, Probability and Statistics, Discrete Math, Computer Architecture, Data Structures & Algorithms, Algorithmic Design, OOP

WORK EXPERIENCE

Amazon

Tempe, AZ

Software Development Engineer Intern

May 2025 – Aug 2025

- Built internal tooling using Java, React, and AWS to find and resolve conflicting product offers for the AmazonCustom team, reducing error diagnosing time by 20% and improving resolution scalability.
- Developing a feature to enable sellers to download sample fulfillment zip files from buyer previews through redux manipulation and new backend functionality, inducing an estimated 22% increase in seller adoption of the Buyer Preview feature and overall seller experience.

Vijay Ganesh Lab

Atlanta, GA

Undergraduate Researcher

Apr 2025 – Present

- Leveraging SAT solvers alongside computer-algebra systems (CAS) to push the boundaries of quantum information science, with an emphasis on characterizing and optimizing Greenberger–Horne–Zeilinger (GHZ) states and solving Krenn’s Conjecture.
- Developing original conjectures in Ramsey theory while advancing SMT techniques for formal verification and stress-testing of cutting-edge AI models and large-scale software systems.

GROWER Lab

Atlanta, GA

Team Lead & Data Engineer

Aug 2024 - Present

- Evaluating different machine learning and threshold-based methods to aggregate power outages in existing data by contrasting with Ground-Truth, partnering with national labs like Sandia and Oakridge to better understand how to approach outage analysis and policy change.
- Leading research to understand the impact of physical and cyber-attacks on US energy infrastructure through data analysis and visualization.

Nationwide Children’s Hospital

Columbus, OH

Data Science Intern

June 2023 – Dec 2023

- Analyzed spatial datasets to identify differences between primary and metastatic tumors, resulting in a 20% faster differentiation which contributed to more targeted cancer treatments.
- Performed various bi-variate correlation studies and created visualizations to understand the architecture of brain tumors.
- Refined an Agent-Based Model's accuracy by 30% by fine-tuning PyTorch parameters, allowing researchers to better simulate cell interactions.

PROJECT EXPERIENCE

GT Marketplace | Typescript, NextJS, ReactJS, Firebase, Python, ExpressJS, NodeJS, HTML/CSS, OAuth2

Apr 2025 – Present

- Designed and implemented a full-stack campus marketplace web enabling students to buy, sell, and trade items with other GT students.
- Includes real-time chats, firebase authentication, and a smart recommendation system for students to dynamically explore the marketplace.

Travel App | Python, Django, Gemini, HTML, CSS, MapBox, Bootstrap, Javascript

Jan 2025 – May 2025

- Created an AI-powered trip-planning engine in Python/Django that combines Google Gemini large-language-model calls with Mapbox routing data to generate fully cost-constrained, day-by-day itineraries, adapting to real-time weather, flight delays, and local events.
- Integrated dynamic recommendation pipelines—budget-filtered hotels, restaurants, and “hidden gems” ranked by hybrid ML + rule heuristics.

Georgia Tech RAG Chatbot | Python, NIM, Docker, ReactJS, PostgreSQL, Scrapy

Aug 2024 – Dec 2024

- Designed a RAG-based LLM in collaboration with AI@GT and NVIDIA, allowing prospective Georgia Tech students/families to easily get accurate information about the institute.
- Built a robust vector embedding pipeline using Scrapy to scrape relevant data and PostgreSQL to store and manage the data.
- Integrated dynamic recommendation pipelines—budget-filtered hotels, restaurants, and “hidden gems” ranked by hybrid ML + rule heuristics.

DeepQuote | C++, Python, PyBind, PyTorch, Gymnasium, Numpy, Stable-Baseline3

Jun 2025 – Present

- Building a high-frequency trading simulator processing 10,000+ orders/sec with order flow analysis, order book mechanics, and other features.
- Using reinforcement-learning-based AI agents to compete in this simulated market, achieving 18-dimensional state spaces and supporting cutting-edge trading strategies with real-time PnL tracking.

SKILLS/INTERESTS

Languages: Python, Java, C/C++, Javascript/Typescript, SQL, HTML, CSS, Assembly

Frameworks: Node.js, React.js, Next.js, Express.js, FastAPI, Django, Flask, React Native, Bootstrap, Dagger, Spring

Software: PyTorch, Transformers, Pandas, Scikit-Learn, PostgreSQL, Firebase, AWS, MongoDB, Docker, Scrapy, Gymnasium, Redux

Awards: Ohio Governor’s Scholar, Hack OH/O 1st place, Huntington Scholar, Presidential Gold Award, Presidential Undergraduate Research Award

Leadership: Events Executive of Trading at GT, President of Taal Tadka Acapella, Team Lead at GROWER Lab, Peer Leader at GT HIVE

Interests: Technology, Basketball, Weightlifting, Cooking, Travel, Poker, Music